

BETWEEN BRAIN AND MIND

NEUROSCIENCE AND COGNITION

Molecular → Cellular → Circuits/Systems

- Neuroscience → how nervous system is organized and works at different levels of investigation
 - Cognition → range of mental processes related to acquisition/storage/manipulation/retrieval of information.
- Neuroscience provides the biological STRUCTURE on which cognition provides FUNCTION.

BRAIN AND AI

- Human brain proves that general intelligence is possible (Biomimicry)
- all-or-none neurons $\approx$ 1/0 artificial neurons

⚠ cells also use biochemical changes

⚠ cells work in cycles rather than chains of causality

IT'S NOT JUST "SOFTWARE" THAT USES "HARDWARE" TO DO THINGS. RATHER THEY BOTH CHANGE WHILE WORKING, ONE IN FUNCTION OF THE OTHER

- Hardware + Software $\neq$ Mind + Brain (first are separable while the second not)
- Human intelligence → generalized learning, limited memory, cycles
- Machine Learning intelligence → statistical learning, vast memory, linear chains of causality

LOCALIZATIONISM vs AGGREGATE FIELD

- Localizationism → faculties are localized in specific regions of cerebral cortex
- Aggregate field → whole brain participated in behaviours.

⚠ They are both implemented by the brain

THE FORM OF YOUR SKULL WILL TELL ME ENOUGH TO REDUCE YOUR PERSONALITY

(PHRENOLOGY) → PSEUDOSCIENCE
Built on approval by using correlational approach

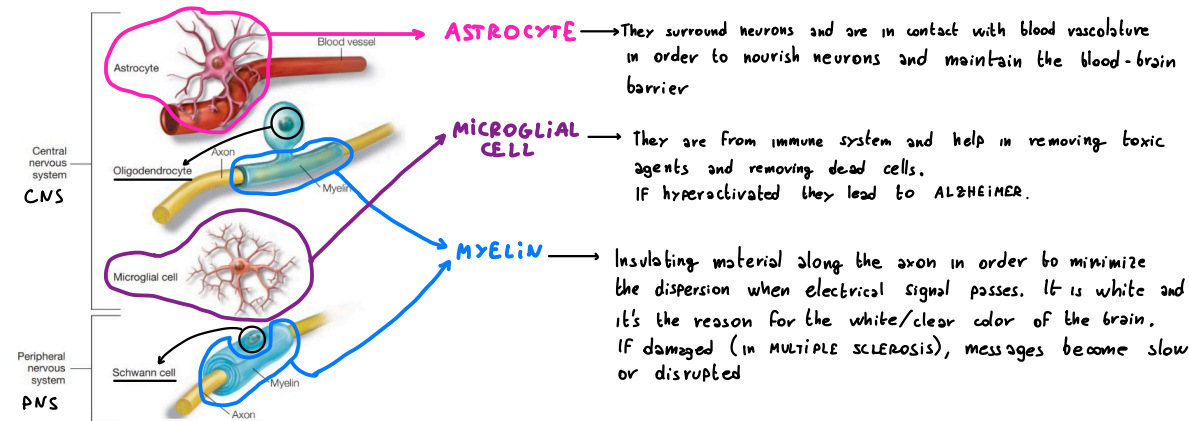
COGNITIVISM

- We can study mental processes because the brain gives rise to them.
- MONTREAL PROCEDURE → stimulation via electrical probes and the following observation of the patient's reaction help us to (still used today) identifying which brain components give rise to which mental processes.

NERVOUS SYSTEM: INDIVIDUAL CELLS

GLIAL CELLS (NEUROGLIA)

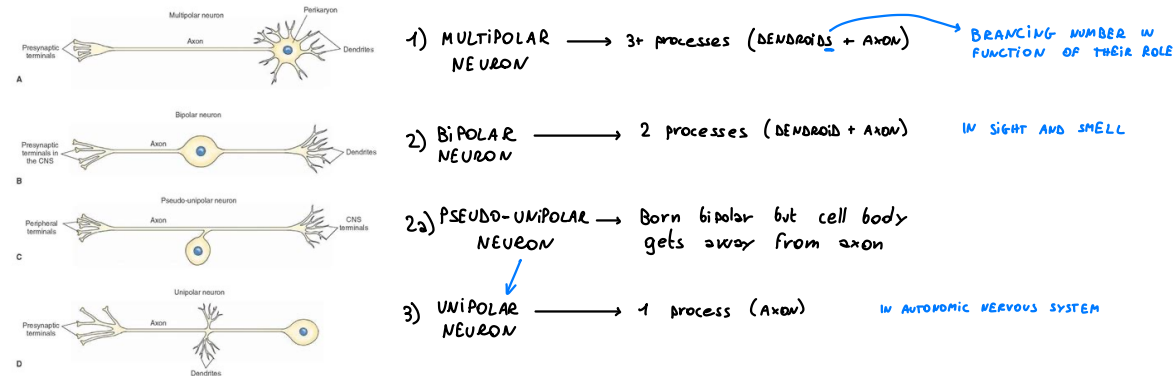
They are the most frequent, outnumbering nerve cells (2-10x) and they are crucial in supporting them.



NERVE CELLS (NEURONS)

△ PROCESSES: any extension that comes out of the cell body! (dendrites or axons)

There are more than 100B of them and around 100 different types, Varying form, location, interconnectivity in relation to their functions:



NEURON STRUCTURE

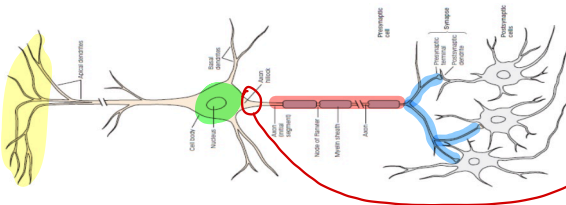
1) GENERAL COMPONENTS:

- Cell membrane $\longrightarrow$ separates extra/intra-cellular fluids
- Cytoplasm $\longrightarrow$ intercellular fluid made by a combo of ions like Potassium (K) or Sodium (Na)
- Extracellular fluid $\longrightarrow$ neurons bath in that, composition similar to cytoplasm
- Cell body/Soma $\longrightarrow$ metabolic centre of the cell

2) **DENDRITES**: Receiving zone of the neuron, there are multiple of them in order to provide multiple inputs to the neuron.

3) **AXON**: Transmitting zone of the neuron, carries output signal for a wide range of distances [0.1 mm, 2m]

4) **SYNAPSES**: Output zone of the neuron, splits the axon signal and delivers it the dendroids of the receiving neurons.



AXON HILLOCK

It is the part where the decision to fire or not is made, if the combined input signals are strong enough

NEURON INFORMATION TRANSFER

1) INPUT ZONE

Signals are graded into amplitude and duration

- 1) **POST-SYNAPTIC POTENTIALS (PSPs)** $\longrightarrow$ From other neurons
- 2) **RECEPTOR POTENTIAL** $\longrightarrow$ physical/chemical signals from environment/cells

2) TRIGGER ZONE

Sums the inputs and generates ACTION POTENTIAL (AP) if the sum $\geq$ threshold (-55mV).

Then higher the sum and higher the frequency of AP.

⚠ **FREQUENCY CODE** $\longrightarrow$ inputs of different amplitudes and different frequencies are converted/coded into constant amplitude and duration and new frequency $\longrightarrow$ CARRIES THE INFORMATION

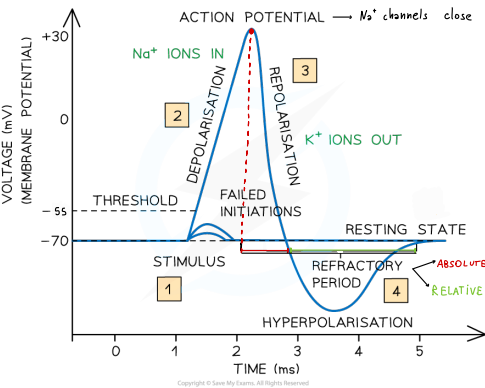
3) CONDUCTIVE ZONE

AP travels as a "all-or-none"/binary signal along the medium.

4) OUTPUT ZONE

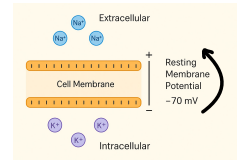
AP reaches **SYNAPSE** $\longrightarrow$ **ELECTRICAL** $\longrightarrow$ signal directly passes to postsynaptic neuron
 $\searrow$ **CHEMICAL** $\longrightarrow$ Frequency of AP $\propto$ NEUROTRANSMITTER

MEMBRANE POTENTIAL "CYCLE"



1) RESTING POTENTIAL

The difference of ions distribution in extracellular and cytoplasm leads to a -70 mV potential difference that can be used by the neuron as battery.



EPSPs (+) inside $\Rightarrow$ cytoplasm is LESS NEGATIVE $\Rightarrow$ "POTENTIAL DECREASES" BUT MATHEMATICALLY INCREASES

Excitatory PSP or EPSPs (+) decreases membrane potential but significant changes are not made by single potentials since PSPs are small in amplitude and passively conducted through cytoplasm and so suffer from DECREMENTAL CONDUCTION (max 1mm).

$\Rightarrow$ SUMMED UP in AXON HILLOCK $\rightarrow$ SPATIALLY: PSPs received from different synapses
 $\rightarrow$ TEMPORALLY: PSPs received at different times

Nothing happens if the total voltage is ≤ -55 mV (firing threshold) and the signals will be considered failed initiations and be ignored, otherwise starts depolarisation.

2) DEPOLARISATION

After reaching the threshold, voltage-gated Na^+ channels open allowing them to enter the cytoplasm from the extracellular and causing a great increase in the membrane potential.

This happens until the Na^+ equilibrium potential ($+30$ mV), then the channels close.

3) REPOLARIZATION

Voltage-gated K^+ channels open allowing them to flow out of the neuron, decreasing membrane potential even below the resting potential, the K^+ equilibrium potential.

In this phase we start to have the ABSOLUTE REFRACTORY PERIOD where no input could cause depolarization as Na^+ channels are close.

4) HYPERPOLARIZATION

The decrease of membrane potential is even ≤ -70 mV (≈ -80 mV), Na^+ channels reopen so we face the RELATIVE REFRACTORY PERIOD where depolarization is possible but with stronger inputs. The K^+ channels close and the IONS PUMP helps to return to resting potential.

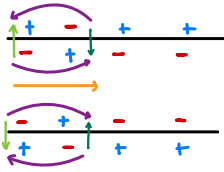
$(3 \text{ Na}^+ \rightleftharpoons 2 \text{ K}^+)$ Actively uses ATP to perform the transport the ions to proper concentration.

HOW INFORMATION (APs) TRAVELS FAR AND FAST?

• FAR

1) THRESHOLD FOR INITIATION → only meaningful information matter

2) NOT DECREMENTAL CONDUCTION → actively propagated, less damaged signal



self-regenerative, so that amplitude stays constant along distances

As the current passes through an axon segment, AP depolarizes the next portion (Na^+ channels open) and regenerates the signal, then hyperpolarizes the previous section, so that AP doesn't flow back.

③

3) REFRACTORY PERIOD → limits frequency of APs (prevent overstimulation)

unidirectional current flow



a) Absolute → right after the AP, impossible to elicit another AP

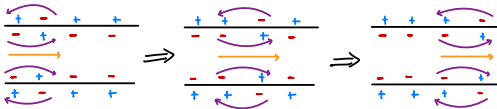
b) Relative → after ③ some K^+ channels close and Na^+ ones recover from inactivation and so it's possible to trigger another AP but with stronger stimuli

4) BINARY NATURE → since duration and amplitude of AP are constant, the signal can be seen as 1/0 and so only frequency gives us information, so we lose less information if the signal is damaged (we just need to detect presence/absence)

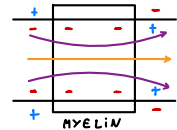
• FAST

→ SALTATORY CONDUCTION

In order for current to be conducted in the medium, it has to change voltages around it.



vs



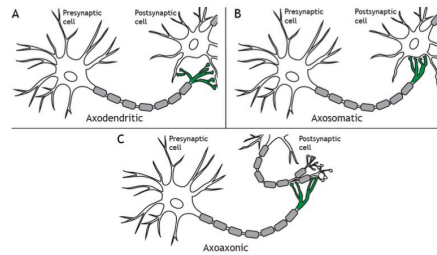
This process has to happen for each particle/channel at time.

With myelin, portions of axon are isolated from extracellular liquid and so the change of voltage occurs only in nodes of Ranvier between myelin making current travel faster.

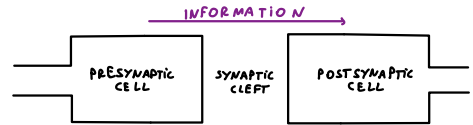
It is saltatory as current seems to jump from a node to the other.

NERVOUS SYSTEM: NEURONS COOPERATION

SYNAPSE: NEURON COMMUNICATION



CONNECTION STRUCTURE



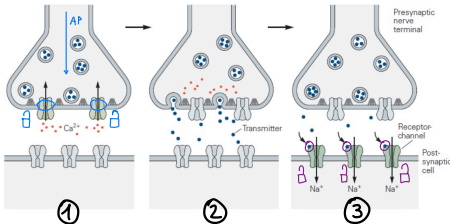
There are 3 types of synapse:

- 1) AXOSOMATIC → connected to postsynaptic-cell's soma
- 2) AXODENDRITIC → connected to postsynaptic-cell's dendrite
- * 3) AXOAXONIC → connected to postsynaptic-cell's axon
 ↳ Inhibition/facilitation role → depressing/enhancing transmitter release over postsynaptic cell

Synapses differ also on the communication medium:

- 1) ELECTRICAL → instantaneous transmission of the flow of ions, Fast, synchronous ISOPOTENTIAL
 ↳ there is a GAP JUNCTION between the 2 cells and ions flows from presynaptic to postsynaptic cytoplasm (cytoplasm continuity) through pores. BIDIRECTIONAL ???
 ↳ When a behavior change is needed (during learning), here we would need to change the whole structure, making it less plastic

- 2) CHEMICAL → diffusion of neurotransmitter across synaptic cleft, slower, allows modulation



- 1) AP reaches presynaptic terminal and opens voltage-gated calcium channels to open, allowing the synaptic cleft's Ca^{2+} to enter.
- 2) Hig. concentration of Ca^{2+} make vesicle fuse with presynaptic membrane, releasing neurotransmitter in the synaptic cleft.
- 3) Neurotransmitter makes the postsynaptic's Na^+ channels to open, so Na^+ can enter and change the potential.

⚠ Effect of neurotransmitter on postsynaptic neuron depends on postsynaptic receptor

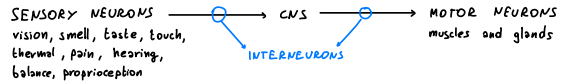
- ↳ This makes the neuron more plastic to change as only the receptor has to change
- ↳ Same input can give totally opposite outputs → HIGHLY SPECIFIC

NEURAL CIRCUITS : COMBINATION OF NEURONS

• No complex behaviour is initiated by a single neuron.

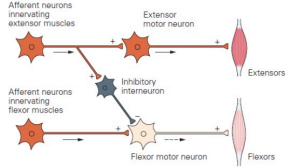
• INTERNEURONS allow:

- 1) PARALLEL PROCESSING with multiple neural pathways for different information of the same stimulus.
- 2) DIVERGENCE: $1 \rightarrow M$, single neuron can have a wide influence on other neurons' activation
- 3) CONVERGENCE: $M \rightarrow 1$, we ensure granularity in neuron activation as there are many contributions (variables) to make it shot



FEED-FORWARD vs FEED-BACK (INHIBITION)

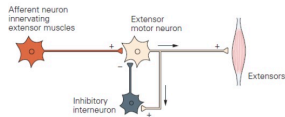
A Feed-forward inhibition



+ excitatory
- inhibitory

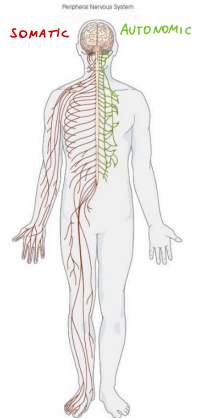
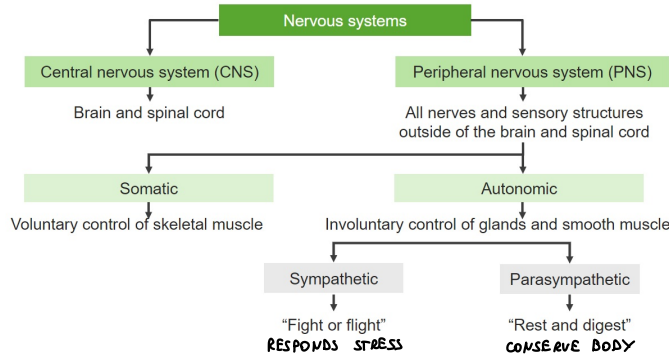
- ① Helps the activation of the current pathway not only by providing high signals to the current pathway's next one but providing low signals to opposite paths, increasing the likelihood of activating it.

B Feedback inhibition



- ② Regulates the current pathway not to exceed a critical level.

NERVOUS SYSTEM



1) PERIPHERAL NERVOUS SYSTEM

Here we have components that perceive stimuli from the extern to the CNS and execute responses (motor/sensory).

- PARASYMPATHETIC → uses acetylcholine, calms down
- SYMPATHETIC → uses norepinephrine, stimulates adrenaline

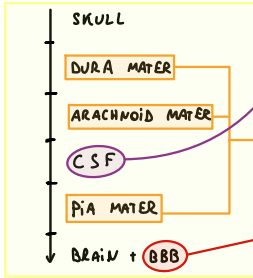
⚠ SYMPATHETIC and PARASYMPATHETIC operates antagonistically on the same organs with opposite stimuli in order to achieve HOMEOSTASIS. BODY OPTIMAL CONDITION

the majority is in common, some are controlled by only one

SOMATIC AUTONOMIC

2) CENTRAL NERVOUS SYSTEM

- Protected by:

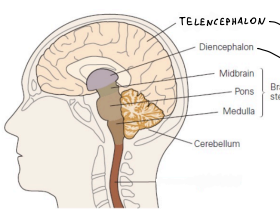


CEREBROSPINAL FLUID → makes the brain float to reduce shocks during accelerations and releases brain's weight

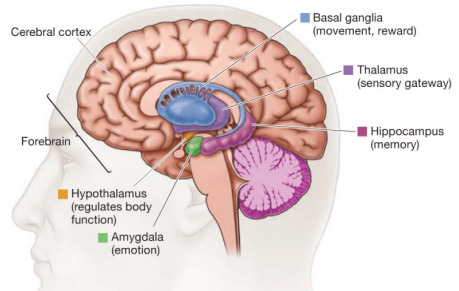
MENINGES → Three layers that cover external layers after skull

BRAIN-BLOOD BARRIER → protects from toxic agents, it is not concentrated but rather diffused around brain

- The brain is divided into:



CEREBRAL CORTEX → FRONTAL LOBE, PARIETAL LOBE, TEMPORAL LOBE, ASSOCIATION CORTEX
BASAL GANGLIA
AMYGDALA
HIPPOCAMPUS
THALAMUS
HYPOTHALAMUS



- BRAIN STEM** → pressure, respiration, sleep
- CEREBELLUM** → posture, coordination, motion
- THALAMUS** → pathway for sensory information
- HYPOTHALAMUS** → regulates autonomic system, rewarding
- AMYGDALA** → determines the stimulus relevance and how to threat it
- HIPPOCAMPUS** → declarative memory, memory formation **⚠ Memory is not monolithic! Long/short-term, declarative or not,...**
- CEREBRAL CORTEX** → surface with many foldings to put neurons in more 3D relationships without using longer axons
 - FRONTAL LOBE** → **MOTOR CORTEX** + **PREFRONTAL CORTEX** (planning, organizing, decision making)
 - M1** (primary motor cortex) → motor homunculus
 - PARIETAL LOBE** → integrates inputs from outside/inside and memory
 - S1** (primary somatosensory cortex) → sensory homunculus
 - TEMPORAL LOBE** → sound processing
 - OCCIPITAL LOBE** → can be activated by many sensory to obtain an integrated experience (high-end abilities) (**ASSOCIATION CORTEX**)
- BASAL GANGLIA** → groups of components, role in motor control and reinforcement learning

Has 2 pathways:

- DIRECT**: disinhibition of thalamus → initiates movement

SWITCH

SUBSTANTIA NIGRA PARS COMPACTA

projects to **striatum**, activates or not a pathway by releasing or not dopamine

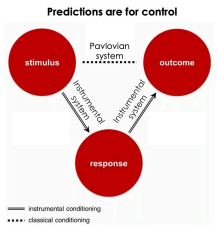
It's the interface where reward influences action, making it more/less likely to occur

depends on

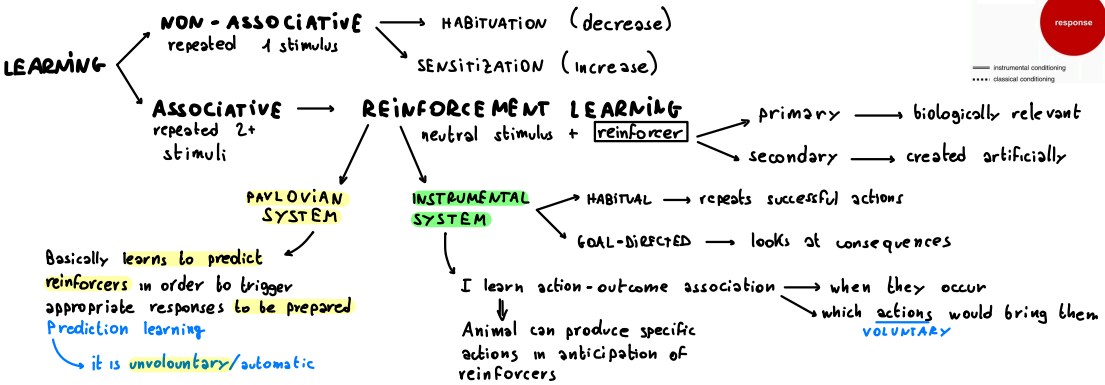
$\text{Reward}_{\text{predicted}} - \text{Reward}_{\text{actual}}$

- INDIRECT**: normal inhibition of thalamus → inhibits movement

ANIMAL REINFORCEMENT LEARNING



LEARNING



PAVLOVIAN LEARNING

In practice we learn stimulus-outcome associations in order to predict when reinforcers occur and which stimuli precede them.

a) → NR

b) US → UR

c) + → UR

d) CS → CR

a) Stimulus with no meaning for the animal → NO RESPONSE (NR)

UNCONDITIONAL STIMULUS (US)

b) Reinforcer stimulus → UNCONDITIONAL (UR) RESPONSE NO LEARNING

c) Reinforcer paired with meaningless stimulus → UR

GENERALIZATION = other stimuli similar to CS can produce CR

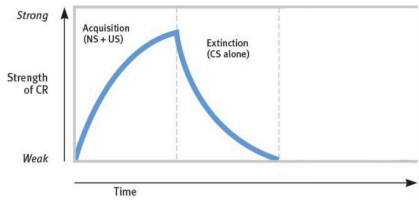
d) Meaningless stimulus → UR LEARNING

now it's called CONDITIONED STIMULUS (CS) now it's called CONDITIONAL RESPONSE (CR)

BETTER SAFE THAN SORRY

It was proven that the conditioned responses are initiated before the unconditioned stimulus, this happens as it allows to be better prepared to respond to what we predict that will happen.

This is not a rigid process but rather flexible along time: ACQUISITION and EXTINCTION OF CS



So we can remove a CR as it is not useful anymore in the current environment, that is as useful as learning from a survival point of view.

⚠️ EXTINCTION ≠ FORGETTING → It's active suppression!

⇒ CS can come back again spontaneously after waiting time or after changing the context in which CS was learned

So we are not just deleting the original one but we are inhibiting the expression of the conditioned response and enforcing a new opposite conditional stimulus.

NEURAL PLASTICITY

It's the ability of neurons to modify connections if required by experience or learning. USE IT OR LOOSE IT

- 1) SHORT-TERM CHANGES → Functional changes to optimize the connection that already exists.
- 2) LONG-TERM CHANGES → structural changes (anatomical alteration) to new connections/synapses or pruning old ones.

⚠ The amygdala is where US triggers UR, so it's where occur changes during Pavlovian learning.

NEURONS THAT FIRE TOGETHER, WIRE TOGETHER

US triggers UR, CS doesn't trigger UR/CR because the connection between CS detector and UR/CR doesn't exist or is too weak. By pairing US+CS we have that CS and UR fire together and they will wire together even stronger.

DOUBLE DISSOCIATION

AMYGDALA → expression of CR (implicit memory)

HIPPOCAMPUS → declarative expression of CS-US (explicit memory)

IF ONE OF THE TWO GETS DAMAGED THE OTHER STILL WORKS AND HAPPENS.
1) DON'T KNOW WHAT'S HAPPENING
2) KNOW SOMETHING THAT DOESN'T HAPPEN

INSTRUMENTAL LEARNING

In practice we learn action-outcome association in order to predict when reinforcers are likely to occur and which action brings them. This allows to anticipate reinforcers rather than reacting to them.

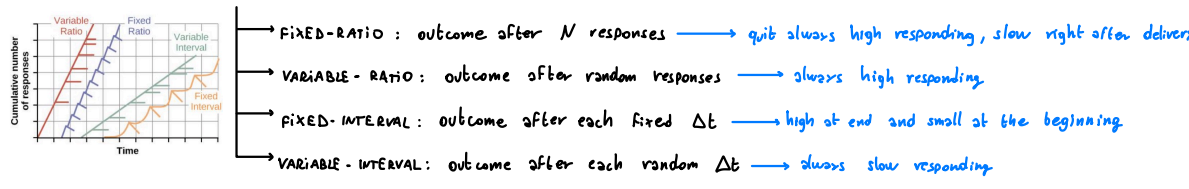
The different reinforcers can be divided by:

- 1) Their effect on the subject

	Delivery	Omission	
Appetitive	Positive reinforcement	Negative punishment	↓ DECREASE BEHAVIOUR
	Negative punishment	Negative reinforcement	↑ INCREASE BEHAVIOUR

- 2) The frequency of their occurrence

- CONTINUOUS REINFORCEMENT: outcome follows behaviour every time it occurs → faster but less robust to extinction
- PARTIAL REINFORCEMENT: outcome doesn't follow behaviour every time it occurs → slower but more robust to extinction



CONTIGUITY, CONTINGENCY, SURPRISE FOR REINFORCEMENT LEARNING

- ONSET
- OFFSET

CONTIGUITY

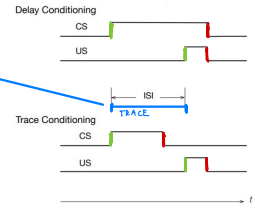
It's the time closeness between stimulus/behaviour and outcome.

We can test it by comparing learning under TRACE and DELAY CONDITIONING.

The last one is better for a better acquired CR. $\Rightarrow$ EXPERIMENT

⚠ NOT SUFFICIENT FOR LEARNING $\rightarrow$ we don't learn 2 random and contemporary events

No learning if it's too long



CONTINGENCY

It's the causal relationship between stimulus/behaviour and outcome. When a stimulus depends on another one they will become associated. $P(US|CS) > P(US|\neg CS)$

⚠ NOT SUFFICIENT FOR LEARNING $\rightarrow$ we don't learn something new if we could expect it to happen



SURPRISE

BLOCKING EFFECT: if it was predictable, it is assumed to not bring anything new to learn

It occurs that we only learn from surprising events, namely event we don't know enough to be well predicted.

The PREDICTION ERROR becomes a very useful measure to look at in order to update expectation and in order to reduce the metric itself the next time.

$\rightarrow$ Evaluates effectiveness of US

COMPUTATIONAL MODELS

The mathematical models for formalize the learning process in finding an effective solution.

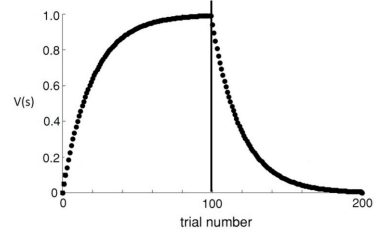
1) RESCORLA - WAGNER (T = n. trials)

- PREDICTION ERROR $\rightarrow$ difference between actual and predicted outcome of US

$$\delta_T = R_T - V_T$$

- EXPECTANCY CORRECTION $\rightarrow$ incrementation/decrementation by the prediction error weighted by learning rate $\alpha \in [0, 1]$

$$V_{T+1} = V_T + \alpha \delta_T$$



$$\begin{aligned} V_0 &= 0 \\ \alpha &= 1/2 \\ R_{0-100} &= 1 \\ R_{100-200} &= 0 \end{aligned}$$

This model predicts the acquisition/extinction curves and the blocking effect.

⚠ Doesn't explain 2nd order conditioning $\rightarrow$ the second CS₂ has no US to be paired with but only the just learned CS₁

⚠ Could we consider CS₁ as US in learning CS₂ ??

2) TEMPORAL DIFFERENCE (T = time steps)

- PREDICTION ERROR $\rightarrow$ also considers the difference between the last 2 predictions (present + past), looking for gain in prediction

$$\delta_T = R_T - (V_{T-1} + V_T)$$

or

$$\delta_T = R_T + V_T - V_{T-1}$$

DOPAMINE IN LEARNING

MODULATES SYNAPTIC PLASTICITY

It doesn't directly cause synaptic changes but it decides when/where/how much they can occur.

DOPAMINE

like a neurotransmitter but with more roles than inhibition/excitation

It's a NEUROMODULATOR which plays a crucial role in the learning process.

Follows those pathways:

- 1) NIGROSTRIATAL (substantia nigra pars compacta → dorsal striatum): production of movement
- 2) MESOLIMBIC (VTA → nucleus accumbens, septum, amygdala and hippocampus): control of emotions
- 3) MESOCORTICAL (VTA → cortex [medial prefrontal, cingulate, orbitofrontal, perirhinal]): decision making

⊆ BASAL GANGLIA
MOTOR LOOP

VENTRAL TEGMENT AREA

! DOPAMINE ≠ REWARD SIGNAL

Dopamine has the objective to minimize prediction error and so to minimize its release and so involves the neurons of the CUE THAT PREDICTS THE REWARD and not on the reward itself.

→ those stimuli/behaviours that precede the behaviour → CS

PHASIC FIRINGS of dopaminergic neurons wrt sign of prediction error:

- ⊕ Increase Firing rate → strengthen involved neurons' synapses
 - ⊖ Decrease Firing rate → don't strengthen involved neurons' synapses
- signed scores → more/less THAN EXPECTED QUANTITY of dopamine

Firing pattern changes → biased or primed circuits
↓
change future behaviours

There are many facets on how dopamine discriminate a reward, implementing the prediction error:

- 1) PRESENCE: if it's unexpectedly delivered (↑) or omitted (↓)
- 2) QUALITY: if it's unexpectedly better (↑) or worse (↓)
- 3) MAGNITUDE: if it's unexpectedly more (↑) or less (↓)
- 4) PROBABILITY: if it's unexpectedly more likely (↑) or less likely (↓) to be obtained
- 5) TIMING: if it's unexpectedly earlier (↑) or later (↓)



NOT ONLY ABOUT REWARD PREDICTION

Until now we looked at dopamine only by the Pavlovian point of view (role in predicting reward).

It was observed that people with low dopaminergic activity learn less quickly.

This means that other than Pavlovian learning (stimulus - outcome) it also influence instrumental one (response - outcome).

DOPAMINE AND DRUGS

Until now we saw dopamine was proportioned to expectations as useful to be prepared and to make decisions, but with drugs (like cocaine) as rewards we get a burst of peaks which are not modulated on expectations but are due to the pharmacological action of the drug.

This hijacks the system and bias the decision making procedure since over time all the other rewards become less and less relevant.

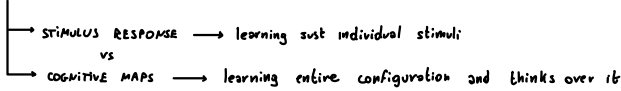
FROM REINFORCEMENT LEARNING TO DECISION MAKING

GOALS AND HABITS

They are part of instrumental learning and we studied them in 4 generations:

1) GENERATION 0

• How do we solve the maze?



- Tested mice on a maze with doors and curtains → the rat was always in a small and blind area
forced the rat to go through them to discover what's behind (from doors)
don't allow the rat to have a long perspective (from curtains)

CRUCIAL FOR LEARNING IN C-S THEORY

Then they used 3 groups of mice wrt how frequent they got reward for solving the maze.

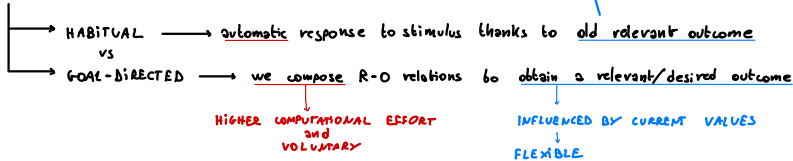
- Resulted that there is LATENT LEARNING that is not shown behaviourally until there is sufficient motivation but rats still responded as if they acquired a mental map of the maze.

ESTABLISHED A DICHOTOMY ON HOW DECISION BEHAVIOR IS CONTROLLED

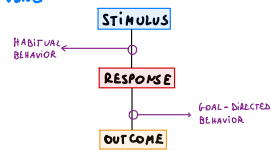
CHALLENGED BEHAVIORISM
PROCESSES MUST BE DIRECTLY OBSERVABLE AND LEARNING IS THE DIRECT CONSEQUENCE OF C-S

2) GENERATION 1

• When we perform a decision, on what does it depend?



This makes them harder to change as they are more inflexible to actual outcome value than past outcome value



• We can answer with a small experiment:

OVERTRAINING: reinforcer used to teach action → **ACQUISITION**

DEVALUATION: reinforced is made less appetible → so we start to enter **EXTINCTION**

TESTING → action performed less → **GOAL-DIRECTED** → **DORSOMEDIAL STRIATUM**
→ action not performed less → **HABITUAL** → **DORSOLATERAL STRIATUM**

THIS BEHAVIORAL DISSOCIATION IS ALSO NEURAL

⚠ BEHAVIOUR IS INITIALLY GOAL-DIRECTED AND REPEATED TRAINING MAKES IT HABITUAL

3) GENERATION 2

• Same objective of previous generation but applied to humans

• Same procedure defined before and measured with fMRI

• It resulted that → **GOAL-DIRECTED** modulated by **MEDIAL ORBITOFONTAL CORTEX** & **MESOCORTICAL DOPAMINERGIC PATHWAY**
→ **HABITUAL** modulated by **DORSOLATERAL STRIATUM** & **MIGROSTRIATAL DOPAMINERGIC PATHWAY**

4) GENERATION 3

• COMPUTATIONAL FORMALIZATION:

GOAL-DIRECTED $\Rightarrow$ model-based

HABITUAL $\Rightarrow$ model-free

MODEL

internal representation of the environment and how it works/responds

Feature	Model-Based	Model-Free
Computational cost	<u>High</u> (needs planning/simulation)	<u>Low</u> (simple retrieval of values)
Flexibility	<u>High</u> (can adapt to changes quickly)	<u>Low</u> (slow to adapt, rigid habits)
Learning speed	<u>Slower</u> (building and updating model takes time)	<u>Faster</u> (direct learning from rewards)
Memory demand	<u>High</u> (stores transitions, outcomes)	<u>Low</u> (stores only action values)
Error correction	<u>Fast</u> (can replay after mistakes)	<u>Slow</u> (needs many experiences)
Sensitivity to changes	<u>High</u> (detects environmental changes quickly)	<u>Low</u> (keeps doing old behaviors)

!!

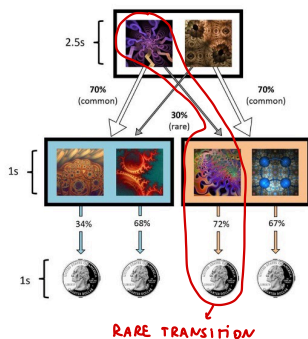
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!!

• MARKOV SEQUENTIAL TWO-CHOICES DECISION TASKS $\longrightarrow$ 2 agents behaves differently under rare transitions

LEARNABLE $\leftarrow$ FIXED OVER TIME

- In each state you can chose a fractal and have a probability to obtain another state in the first case or a reward in the second
- MODEL-FREE $\longrightarrow$ learns the first move even if the reward comes by a rare transition and so not reliable to learn a good strategy, so the learn will be confused and inefficient
- MODEL-BASED $\longrightarrow$ understand rarity of situation and learns it properly



- Result on humans showed that the 2 agents are not that much separated but rather integrated, suggesting a DYNAMIC INTER-DEPENDENCY

OBJECT RECOGNITION: VENTRAL VISUAL CORTEX and DEEP NETWORKS

VISION

ability to capture a useful
image of the environment

ability to use images to understand the
environment and act over it

- Complex procedure that includes both the **INFORMATION-PROCESSING TASK** and **INTERNAL REPRESENTATION** of the image itself.
- **MARR'S TRI-LEVEL HYPOTHESIS**

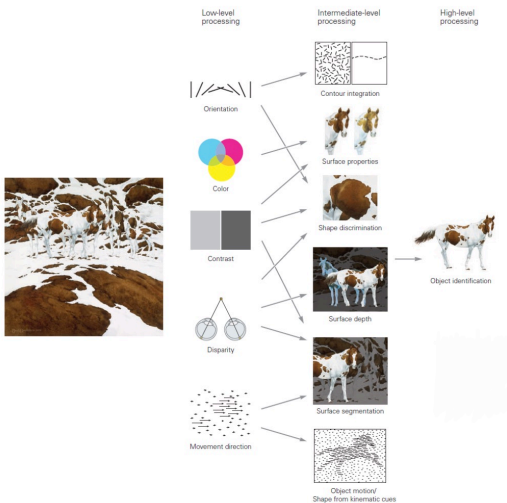
- 1) **COMPUTATIONAL** → which is the goal and the logic of the strategy to obtain it
- 2) **ALGORITHMIC** → how to implement ①? Define input, procedure and output
- 3) **IMPLEMENTATIONAL** → Hardware and/or physical implementation

GENERAL
↓
PRECISE

- ⚠ **VISION ≠ CAMERA** → camera represents point by point in one plane the visual field → camera is more accurate on representation
→ visual system also separates distinct components (eg. background and foreground)
→ identifies moving objects in extreme conditions (light, occlusion, ...) and very fast

- It's a generative process that involves **RETINA** + **EXPERIENCE** + **BUILTIN NEURONAL WIRING**
↓
Bayesian's priors, more important as the image is ambiguous

- Visual scene analyzed in 3 levels of processing: **LOW**, **INTERMEDIATE**, **HIGH**

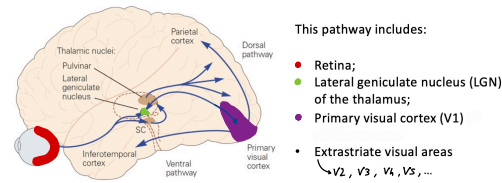


V1

- 1) **LOW** happens in Retina or First Visual Cortex and separates atomic info like color, orientation, contrast, ...
- 2) **MEDIUM** uses atomic info to characterize more general properties of the object
- 3) **HIGH** uses the detected properties to classify the object

RETINO-GENICULO-STRIATE PATHWAY

Mediates visual processing thanks to some components: RETINA + LGN + V1



1) RETINA

Transform the visual input in neural signals (**PHOTOTRANSDUCTION**), then the signal travels through the optic nerve.

- ⚠ **RECEPTIVE FIELD of a neuron** → region of the visual scene where a stimulus excites the neuron
- the region shape/size varies with location in visual system hierarchy
 - The neurons with close receptive fields are close in order to maintain spatial relationships.

Light arrives already processed by lenses and hits the back of the eye where it finds the retina that processes it with some steps:

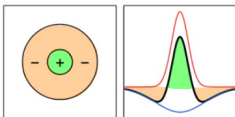
- 1) **PHOTORECEPTORS** = $\overset{\text{Intensity}}{\uparrow}$ **RODS** + $\overset{\text{colour}}{\uparrow}$ **CONES**, they hyperpolarize/depolarize in presence/absence of light producing a simple signal representing the input
- 2) **BIPOLAR CELLS** → simply acts like a connection between ② and ③ (with also other cells)
- 3) **RETINAL GANGLION CELLS (RGC)**, those neurons have concentric circular receptive fields divided into
 - ON-center cells, which are excited by light in the center of receptive field.
 - OFF-center cells, which are excited by dark in the center of receptive field.

⚠ The two circles are antagonist and so the highest response is obtained by not having light in $\ominus$ areas → **are preferred to** $\ominus$ **only at intermediate frequency**

This happens because in this way we are detecting edges which are a lot more informative.

In this way we are not orientation sensitive: $\text{ON} = \text{OFF} = \text{ON} = \text{OFF}$

⊛ DIFFERENCE OF GAUSSIANS



The 2 circles concentric circles are implemented as difference of 2 gaussians → narrow center → sensible to central light

→ wide base → sensible (negatively) to border light

Then the information are transmitted through the optic nerve. At a certain point there is the optic chiasm between the 2 optic nerves, beyond that point the nerves of both eyes are merged and connected to both hemispheres.

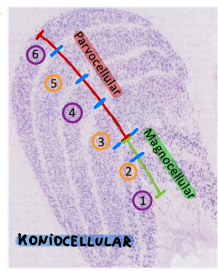
↓
There is a unique channel with 6 layers/channels (3 per eye)

2) LATERAL GENICULATE NUCLEUS (LGN)

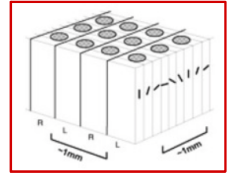
△ THERE IS ONE OF THEM PER EACH EYE THAT MIXES INFORMATION FROM BOTH OF THEM } → CONTRALATERAL EYE
IPISILATERAL EYE

Receives the 6 layers after optic chiasm, they are stored in a pile structure, implementing a spatial register:

- MAGNOCELLULAR, layers capture achromatic contrast information
- PARVOCELLULAR, layers capture r/g contrast information
- KONIOCELLULAR, layers capture b/y opponent information



It has the same RF as in the RGC but in addition performs some operations before passing them to the V1. It takes into account multiple sources, decides what is relevant, then selects it and synchronizes it in order to facilitate V1 in finding relevant information.



3) PRIMARY VISUAL CORTX (V1)

It receives the 6 layers from both LGNs and forms a visuotopic map of the visual field.

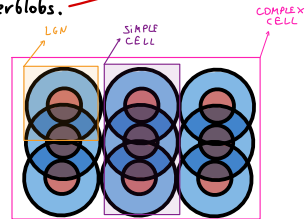
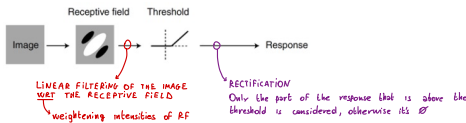
It has the smallest receptive field in the visual system as the first 10° of visual field are handled by more than half of the V1, providing the highest spatial resolution of the center.

Neurons with similar functions are close together in columns (FUNCTIONAL MODULES):

- ORIENTATION COLUMNS, group neurons with the same orientation preferences. From column to column there are changes of 10/15°. With 12 column we complete the sequence of orientations (HYPER COLUMN). The orientations are displayed sometimes like pinwheels that show a circular/radial change (that can be wide) in the orientations.
- OCULAR DOMINANCE COLUMNS, they keep the 2 eyes separated alternating each column to a specific eye.
- BLOBS, located in superficial layers and with circular shape (cylindric columns) that focus on the color
- INTERBLOBS, located around blobs and more focused on orientation
- ICE CUBE MODEL: In 1mm³ we have 2 hypercolumns (one per eye), blobs and interblobs.

We can find 2 types of cell:

- SIMPLE → elongated receptive field and respond to a narrow range of orientations



- COMPLEX → rectangular receptive field in order to be more sensitive to orientation, when put together they are great in detecting movement. Moreover they are less selective for the position in the RF even with inverted contrast, meaning that they are detecting edges. INDEPENDENTLY FROM

Here the response follows the same workflow of simple cells but for N of them in parallel and then the response is given by the sum of the simple responses.

△ SIMPLE CELLS ARE NOT RANDOM → must have same orientation in RF → in order to be orientation sensitive
↓
must have slightly different positions in RF → in order not to be position sensitive

- They are organized into an feedforward hierarchical model of the RF: LGN $\rightarrow$ SIMPLE $\rightarrow$ COMPLEX

⚠ NOT COMPLETE $\rightarrow$ CROSS-ORIENTATION SUPPRESSION

If we consider gratings we will notice that they activate the complex neuron when they match its preferred orientation. But if we consider the same gratings but in the orthogonal direction (shouldn't activate the complex neuron), and we superimpose them to the original gratings we'll notice a lower response (rather than 0). So the last ones are inhibiting even if they shouldn't be detected.

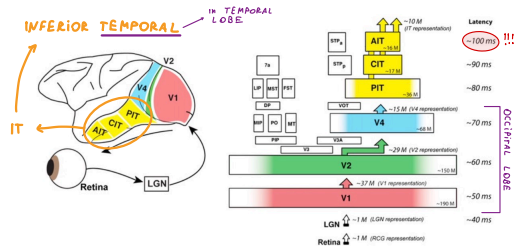
This happens because maybe they are not detected by that specific neuron but maybe from the ones next to it, leading to suppression $\rightarrow$ non-linear procedure

OBJECT RECOGNITION

- OBJECT $\rightarrow$ shape + colour + texture (integrated with perceptions)
- VISUAL RECOGNITION $\rightarrow$ assign a label to an object
 - $\rightarrow$ IDENTIFICATION $\rightarrow$ recognizing the exact instance of an object (simpler for computers)
 - $\rightarrow$ CATEGORIZATION $\rightarrow$ recognizing the class of that instance object (simpler for humans)
- The recognition must be efficient even if we will meet infinitely variations of the image each time, suggesting the need of generalizing, so balancing between:
 - 1) SELECTIVITY $\rightarrow$ different responses to different objects (eg. distinguish 2 faces)
 - 2) INVARIANCE $\rightarrow$ same response to all the transformations of the same object (eg. rotation of face)
- It can be evaluated with:
 - 1) CONSISTENCY $\rightarrow$ how predictions match in various measurements $\rightarrow$ if 2 set of predictions are consistent and with the same accuracy then there are the same correct ones and wrong ones
 - 2) PERFORMANCE $\rightarrow$ accuracy (= correct / total)

VENTRAL PATHWAY (object recognition)

- It is the one on which we'll focus but there is also the DORSAL one which is interconnected and shares information obtained during its computations.



Aspect	Ventral Pathway ("What")	Dorsal Pathway ("Where/How")
Function	Object recognition, identification	Spatial awareness, movement coordination
Pathway	Occipital → Inferior Temporal Cortex	Occipital → Posterior Parietal Cortex
Main Input Type	Parvocellular (local, color)	Magnocellular (global, motion)
Processes	Shape, color, textures	Motion, depth, position
Consciousness	Linked to conscious perception	Often unconscious, guides real-time action
Damage Effects	Visual agnosia (can't identify objects/faces)	Optic ataxia (impaired visually guided movement)
Example	Recognizing a face	Grasping a moving object

1) V2

→ INFERRED STRUCTURE

Allows to see illusory contours, going beyond the simple illumination contrast.

Moreover, when considering an edge, here we decide which side of it belongs to the object.

Finally, it is strongly involved into statistical regularities (eg. surface segmentation).

Less precise RF.

BORDER OWNERSHIP CODING

2) V4

Plays a role in FIGURE-GROUND SEGMENTATION by focusing on curvature and color.

It also deals with partially occluded shapes.

Even less precise RF.

3) IT

From now on the RF is so big that we won't consider it anymore as we consider bigger features.

Here we form a representation of the object → view dependant/independent responses

single (grandmother cell) vs distributed coding → better explains confusion of similar images
USED → better explains human ability in greatly recognize new objects

Some neurons correlate with non-vision task, highlighting the function not only of perception but also of meaningful representations.

* RESPONSE VECTOR → A population of N neurons maps the representation of an object into N -dimensional vectors, in the space created then recognition is about finding separation among representations.

DNN FOR OBJECT RECOGNITION

HIERARCHICAL CONVOLUTIONAL NEURAL NETWORKS (HCNNs)

They were chosen for biological plasticity obtained extending classical CNNs with:

- 1) MIXTURE MODELS → increase breadth complexity to allow processing of fine and coarse feature simultaneously, like in parallel pathways.
- 2) HIERARCHICAL MODULAR OPTIMIZATION (HMO) → architecture optimization without backpropagation (no gradients, YES local learning). Instead it searches over the space by using optimization techniques inspired by evolution.

The process is divided into:

- a) MODULAR SEARCH → divide network in modules and optimize each of them (from lower to higher) with TPE. ④
- b) BOOSTING → we select modules that make diverse error → work on different aspects
- c) COMBINATION → we combine the selected modules in HCNN
- d) REWEIGHTING → when inputs are misclassified we increase their weight (occurrence of samples) for the next learner ⑤
- e) TPE PROBABILISTIC MODEL → identifies which regions of hyperparameter space would improve performance, then it doesn't modify them in previous modules but creates new complementary modules

⇒ LOCAL LEARNING ④ + NO SYMMETRIC WEIGHTS + ADAPTIVE FOCUS ⑤

when CNN don't take time where it's needed, accuracy drops

TAKES TIME

used to select stored knowledge to "complete" the inline input (recalling important information)



FUNDAMENTAL FLOWS OF DNNs

- 1) ABSENCE OF RECURRENT PROCESSING → most CNN are feedforward but visual system adopts also backward loops
- 2) LACK OF ATTENTION MECHANISMS → primates focus on priority regions of the image
- 3) LEARNING MECHANISMS → primates use more unsupervised/self-supervised or reinforcement learning
- 4) TOO "BRUTE FORCE" APPROACH → because of physical constraints, the brain uses sparse and noisy signals efficiently
- 5) REPRESENTATION DIFFERENCES → the fact that CNN use layers like the brain doesn't mean the same representation

RECURRENT NEURAL NETWORKS (RNN)

Recurrent loops are highly presents in the primate ventral visual stream, allowing finer and better performances for non-early neuronal responses (where feedforward nets fail). Those responses are bounded to the most difficult task which are solved since it allows an iterative refinement process of the representation.

IMAGE COMPLETION IN RNN

In animals, pattern completion tasks are implemented by recurrent computations. When the object misses important details it is necessary to take time and iteratively reconstruct it using also other connections.

Moreover recurrent architectures require fewer parameters, flexible FLOPs and can interact with feature changes over time.

UNSUPERVISED NEURAL NETWORK

Humans are not exposed to as many supervised data as NN. On the other hand humans are exposed to temporal related images wrt the independent ones in ImageNet.

A potential method is CONTRASTIVE LEARNING, where we pick an image, we augment it and then we learn that the 2 images are the same (knowing that they are different from all the other images in the batch). This helps in ignoring irrelevant noise and extract the core features of the objects.

The results of comparing unsupervised and supervised learning highlighted a threshold between richness of learning and specialization (specialization vs generalization).

PARTIAL SUPERVISION

As it happens with children maybe they obtain most of information from continuous inputs of real world and practice unsupervised learning, but then we have to mimic the parenting role by giving a relatively small amount of labels. The results with a small portion (3%) of labels massively boost the model's performance.

BACKPROPAGATION AND BRAIN

SYNAPTIC PLASTICITY

LONG-TERM POTENTIATION (LTP) → increase of synaptic strength between pre- and post-synaptic neurons that

activate repeatedly and near-synchronously

→ INPUT SPECIFIC → LTP occurs only in active synapses and doesn't involve inactive neighbours

→ ASSOCIATIVE → if a strongly activated neuron borders with a poorly activated one, also the second one undergoes

LTP (COINCIDENCE DETECTION FEATURE)

PROBLEMS WITH BACKPROPAGATION

There is not direct evidence of the brain using backprop algorithms.

1) NO SUPERVISION SIGNAL → the brain relies more on weak + distributed + local signals guiding learning

2) ARTIFICIAL NEURON → the first returns continuous values while the second one uses AP which is all or none
BIOLOGICAL NEURON

NO BACKPROPAGATION ← NO GRADIENT

3) SIGNED ERROR SIGNAL → in the brain exist some signals (eg. in cerebellum) but it's not clear how they are propagated in deeper networks

4) DIRECTION SYMMETRY → feedforward neurons must have similar strength to their respective feedback neurons

WEIGHT TRANSPORTATION PROBLEM

recent studies show that also random and fixed weights together nudge the network in the right direction → BECAUSE IT IS STILL COHERENT

5) LOCAL ERROR REPRESENTATION → brain neuron relies only on local information about neighbours or local chemical. In artificial neuron the backpropagation brings information about all successors by gradient.

6) FEEDBACK IN INFERENCE → in ANN backpropagation is active during training and latent during testing, while in the brain they are constantly used.

NEURAL GRADIENT REPRESENTATION BY ACTIVITY DIFFERENCES (NGRAD)

The idea is to estimate gradient errors with local activity differences, which are defined like the difference between the forward and backward activity in that neuron.

With this idea we get rid of some crucial problems → 2, 4, 5

DEEP REINFORCEMENT LEARNING

REINFORCEMENT LEARNING

It's a problem where one agent embedded in an environment, improves its actions in response to environmental states.

⚠ RL ≠ SUPERVISED LEARNING → the agent doesn't receive explicit feedback indicating correct actions

DEEP RL

- DEEP → we talk about deep RL when we use a DNN to use self-learned representations → NN as agent
- SAMPLE INEFFICIENCY → they need more samples wrt humans to achieve the same level of performance

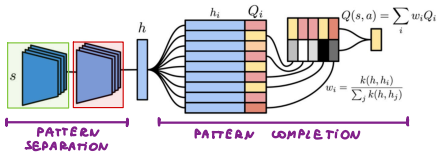
MITIGABLE BY

caused by:

- 1) INCREMENTAL PARAMETER ADJUSTMENT → models slightly adjust the parameters in order not to lose generalization but this cost them a lot of time (as the complexity of the environment grows)
- 2) BIAS/VARIANCE TRADE-OFF → more biases → more quick learning
→ less generalized → less accurate

- EPISODIC DEEP RL → store and reuse explicit records of past events as comparison models

→ NEC (Neural Episodic Control)



- 1) ENCODER → CNN that encodes the pixel image → represent state
- 2) DICTIONARY → store the value of each state
- 3) FINAL NETWORK → estimates the value of the new state looking at similar ones in the dictionary

BACKPROPAGATION → slow training → better representation

⚠ ≠ FROM HUMAN → The 3 components don't work in parallel and don't cooperate to improve themselves

IN THE BRAIN

There are 2 intelligent systems: $\rightarrow$ better captures statistical regularities across experiences

1) NEOCORTEX $\rightarrow$ slow acquisition of structured knowledge about the environment

2) HIPPOCAMPUS $\rightarrow$ rapid learning of specific episodic events

% AUTO ENCODER

CHANGES HOW THE KNOWLEDGE IS STRUCTURED

So initially memories are stored into ② and the more they are replayed and the more changes are caused to ①, integrating the new knowledge.

In the hippocampus we find two core operations:

1) PATTERN SEPARATION $\rightarrow$ make similar inputs more distinct $\rightarrow$ avoid interference

2) PATTERN COMPLETION $\rightarrow$ From partial inputs retrieve complete memories

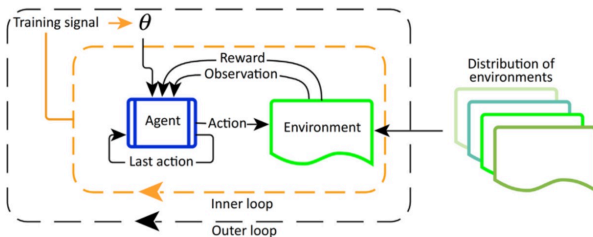
$\rightarrow$ ATTRACTORS $\rightarrow$ consequence of hebbian plasticity, patterns enforce connections between neurons as they usually fire together. These links can be fully activated even when a portion of the pattern is activated, leading to completion.

REPLAY (in hippocampus) $\rightarrow$ reactivation of neural activity patterns, they happen in a compressed timescale in order to invoke memory consolidation in circuits beyond the hippocampus.

META-RL (LEARNING TO LEARN)

Biological learners never start learning from a tabula-rasa situation but always bring past learning on any new learning problem. We tend to learn higher level invariant structures, like intuitive physics, that can provide us powerful bases to quickly learn general structure of tasks and then specialize in a task.

It is important to notice that META-RL emerges spontaneously from more basic mechanisms, it only requires short-term memory and a distribution of interrelated tasks.



• INNER LOOP = CLASSIC RL
 $\rightarrow$ fast adaptation over one present task

• OUTER LOOP = META RL
 $\rightarrow$ slow adaptation $\rightarrow$ optimizes parameter over many future tasks

So we can rethink of sample inefficiency as the missing of the right architecture and the right inductive biases. We have shaped our architecture and biases over evolutionary time (millennia). Then we face lifetime learning (decades) for learning general tasks and only then we have real time (seconds) learning for very specific tasks. So we look faster because we already faced the most time-consuming layers.